AI for Multimodal RGB-Thermal Pedestrian Detection through Efficient Fusion Strategies

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Abstract

Pedestrian detection is an essential computer vision task for autonomous vehicles, surveillance, and search-and-rescue systems. RGB cameras provide rich visual information but perform poorly under low-light conditions, whereas thermal cameras detect heat signatures independent of lighting but lack texture details. This work proposes an **Adaptive Fusion Module** that intelligently combines RGB and Thermal features using **Channel Attention, Spatial Attention, and learnable fusion weights**. The proposed method is evaluated on the **VTUAV-det** dataset and achieves a **5.7% improvement in overall mAP** and **16.3% improvement in small-object detection** over the baseline **QFDet**, while maintaining **real-time performance (17.2 FPS)**.

1. Introduction

Reliable pedestrian detection is critical for applications such as autonomous driving, drone surveillance, and disaster response. RGB images provide detailed appearance information but fail in poor lighting, whereas thermal images remain reliable at night but lack fine texture. Combining both modalities enables robust detection across diverse environments.

The main challenges addressed in this work are:

1. Detecting **small pedestrians** from drone imagery.
2. Handling varying reliability of RGB and Thermal data under different lighting conditions.
3. Maintaining accurate RGB-Thermal feature fusion.
4. Achieving real-time inference with minimal computational overhead.

Our objective is to design an efficient adaptive fusion strategy that improves pedestrian detection while preserving computational efficiency.

2. Dataset and Baseline Analysis

2.1 Dataset

Experiments were conducted on the **VTUAV-det** dataset, which contains paired RGB-Thermal drone images with pedestrian annotations.

Split	Images	Pedestrian Instances
Training	1,200	8,138
Validation	300	2,337
Test	200	2,068

2.2 Pedestrian scale distribution:

1. Small (<32^2 pixels): 42.3%
2. Medium (32^2–96^2 pixels): 35.7%
3. Large (>96^2 pixels): 22.0%

The large proportion of small pedestrians makes tiny-object detection the primary challenge.

2.3 Baseline Performance

Model	mAP	APs	FPS
RGB Only	0.452	0.213	18.5
Thermal Only	0.423	0.286	18.2
QFDet Baseline	0.487	0.325	16.8

The experiments show that RGB performs better on large objects, while Thermal improves small-object detection. This complementary behavior motivates multimodal fusion.

3. Proposed Adaptive Fusion Strategy

To improve RGB-Thermal feature integration, we introduce an **Adaptive Fusion Module** consisting of four components:

3.1. Channel Attention

Channel Attention emphasizes informative feature channels using global average pooling followed by fully connected layers.

[A_c(F)=\sigma(FC(GAP(F)))]

This suppresses less useful channels while highlighting important RGB and Thermal features.

3.2. Spatial Attention

Spatial Attention identifies important image regions through average pooling, max pooling, and a 7×7 convolution.

[A_s(F)=\sigma(f^{7\times7}[Avg(F),Max(F)])]

It improves localization of pedestrians while reducing background interference.

3.3. Adaptive Weighted Fusion

Instead of assigning equal importance to both modalities, a learnable parameter dynamically adjusts their contribution.

[F_{fused}=\alpha F_{RGB}+(1-\alpha)F_{Thermal}]

When lighting is good, the network relies more on RGB features; during low-light conditions, greater importance is automatically given to Thermal information.

3.4. Feature Refinement

The fused features are refined using convolution and residual connections to improve feature representation before detection.

Training was performed using SGD with momentum (0.9), learning rate 0.001, batch size 2, and 12 training epochs.

4. Results and Discussion

4.1 Detection Performance

Model	mAP	APs	FPS
QFDet Baseline	0.4867	0.3245	16.8
Proposed Model	0.5150	0.3780	17.2

4.2 Performance Improvements

1. Overall mAP: +5.7%
2. Small-object AP: +16.3%
3. Medium-object AP: +4.9%
4. Large-object AP: +3.0%
5. Inference Speed: Improved to 17.2 FPS

The proposed model increases parameters by only **3.1%**, demonstrating that significant accuracy gains can be achieved with minimal computational cost.

4.3 Ablation Study

Configuration	mAP	APs
Baseline	0.4867	0.3245
+ Channel Attention	0.4980	0.3420
+ Spatial Attention	0.5050	0.3580
+ Adaptive Fusion	**0.5150**	**0.3780**

The adaptive fusion mechanism contributes the largest improvement, while Channel and Spatial Attention provide complementary performance gains.

5. Conclusion

This work presents an efficient **Adaptive RGB-Thermal Fusion Module** for pedestrian detection in drone imagery. By combining **Channel Attention**, **Spatial Attention**, and **learnable adaptive fusion weights**, the proposed approach effectively exploits complementary information from RGB and Thermal modalities.

Experimental results on the VTUAV-det dataset demonstrate:

1. 5.7% improvement in overall mAP
2. 16.3% improvement in small-object detection
3. Real-time performance (17.2 FPS)
4. Minimal increase in model complexity

The proposed method is suitable for practical deployment in autonomous driving, intelligent surveillance, search-and-rescue missions, and other applications requiring reliable pedestrian detection under varying environmental conditions.

References

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